

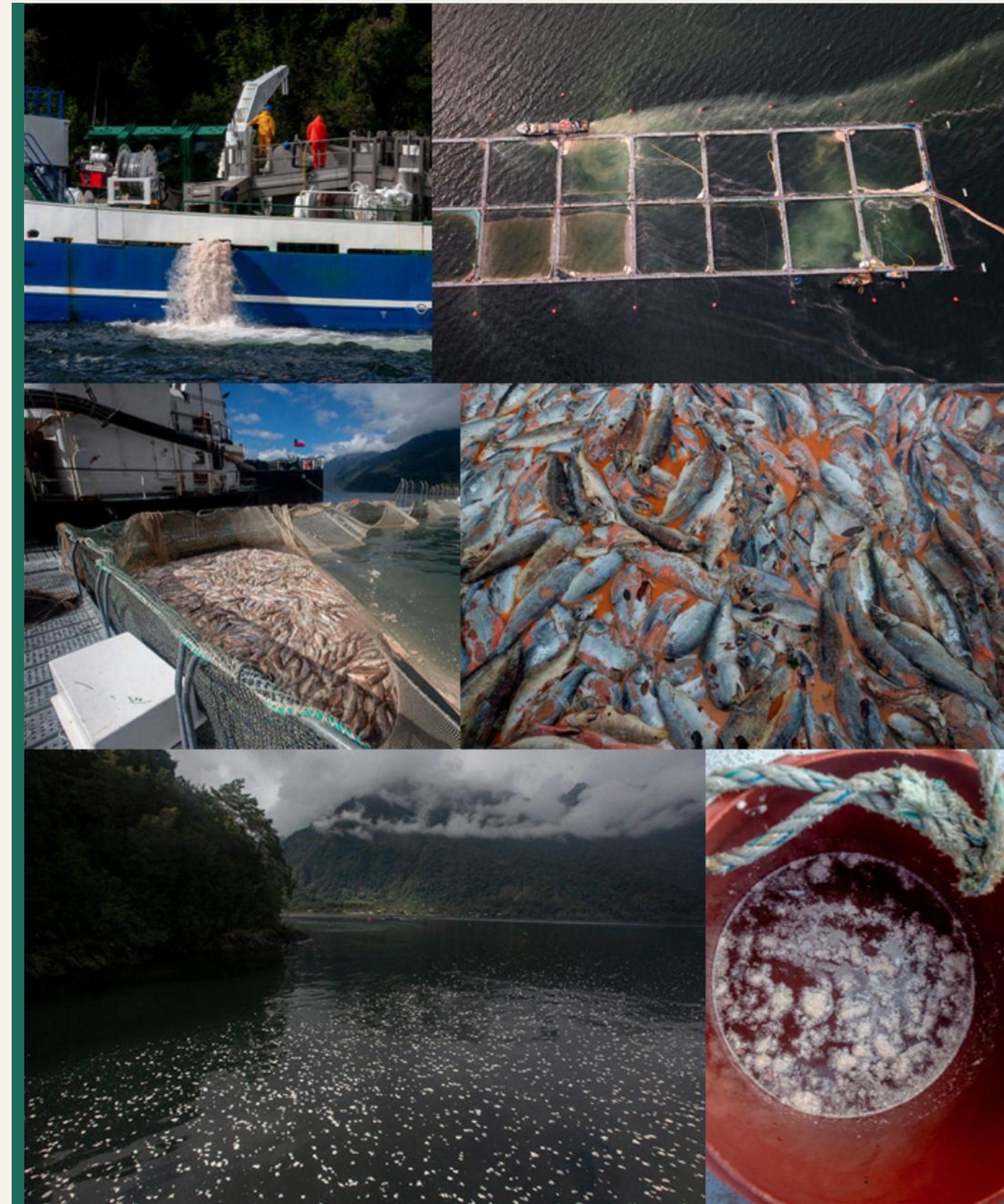


Figure 1. Photographies taken from Comau fjord in southern Chile showing the dispossal and propagation of salmon industry waste directly to the enviroment. Taken by Alvaro Vidal (2021).

What are the impacts of intensive animal husbandry on biological systems?

The human population has more than doubled over the past six decades, imposing an unprecedented demand on global food systems.

This has biological consequences on the organisms it exploits and the surrounding ecological communities.

Genetic approaches applied to major domesticated species have revealed genome-wide signatures of domestication and breed development, evidence of selective sweeps and demographic bottleneck events; intensive artificial selection compresses evolutionary change into timescales of decades rather than millennia.

Prolonged and intensive animal farming creates conditions for the emergence of epidemics, increasing the risk of pathogen emergence and zoonotic spillover.

Studying such link between production intensification, genomic dynamics, and pathogen unbalanced ecology demands an integrative analytical lens within a One Health framework.

Aquaculture represents the food production sector where these dynamics converge with particular acuity.

As the world's second-largest producer of farmed Atlantic salmon (*Salmo salar*) and coho (*Oncorhynchus kisutch*), Chile concentrates all of these dynamics into a single tractable study system.

In Chile the industry has over four decades of continuous large-scale production, during which multiple epidemiological crises have occurred providing natural experiments in pathogen-driven selection, and naturalized anadromous salmonid populations have established, constituting genuine wild comparators.

By characterizing genome-wide diversity, population structure, and demographic history of cultivated salmonids in southern Chile alongside signatures of selection associated with domestication and disease pressure, and by coupling these analyses with a temporal reconstruction of microbial, viral, and eukaryotic community shifts in the surrounding environment, this project aims to elucidate how large-scale, long-term aquaculture reshapes genetic and pathogenic processes at both the organismal and ecosystem levels in southern Chile salmon farms.

Methods

Database data acquisition

(NCBI/ENA/SERNAPESCA)

WGS and MHC locus data of *Salmo salar*, *Oncorhynchus mykiss*, and *Oncorhynchus kisutch* will be downloaded if it fulfills minimal metadata requirements. Also reference genomes.

Metagenomic water datasets from salmon aquaculture, focusing on 16S, 18S, and ITS markers, will be downloaded under the same condition.

Lonitudinal epidemiologic data of *Orthoreovirus piscis*, Infectious pancreatic necrosis virus and *Piscirickettsia Salmonis*.

Three historic cohorts have been defined: 2010-2014, 2015-2019, 2020-2025.

Sampling and sequencing

Three sampling sites will be selected within the Los Lagos region of Chile corresponding to the most productive farms of the region.

Blood samples from cultivated (n>80) and naturalized specimens will be collected. Water and sediment samples will also be collected.

Extracted DNA will be sequenced using the Illumina NextSeq2000 2x151 and Oxford Nanopore MinION for long read data.

Bioinformatic analysis

All bioinformatic analyses will be done in a cluster running Ubuntu Linux, with Conda manager.

Obtained sequences will be curated as necessary, using Trimmomatic and Prinseq++. FastQC/MultiQC will be further used to check post-curation quality. For Nanopore output, a Nextflow pipeline will be used.

Sequences will be mapped using BWA against NCBI references for each species. Duplicate removal will then be performed with Picard/SAMtools-markdup. Nextflow variantcatalogue pipeline will be applied for variant calling analysis. Pipeline output will serve mainly for PCA, admixture proportions, demographic history, allelic trajectories and funtional annotation. Final results will be stratified by historic cohorts and correlated with epidemiologic data in R.

As for metabarcoding analysis, every marker will follow the same generalized pipeline: Cutadapt for demultiplexing samples, DADA2 (R) for ASV inference, merge of pairs and Taxonomic assignment based on databases created with CRABS. PCoA and alpha diversity measures will be performed and contrasted with the aforementioned epidemiological data.

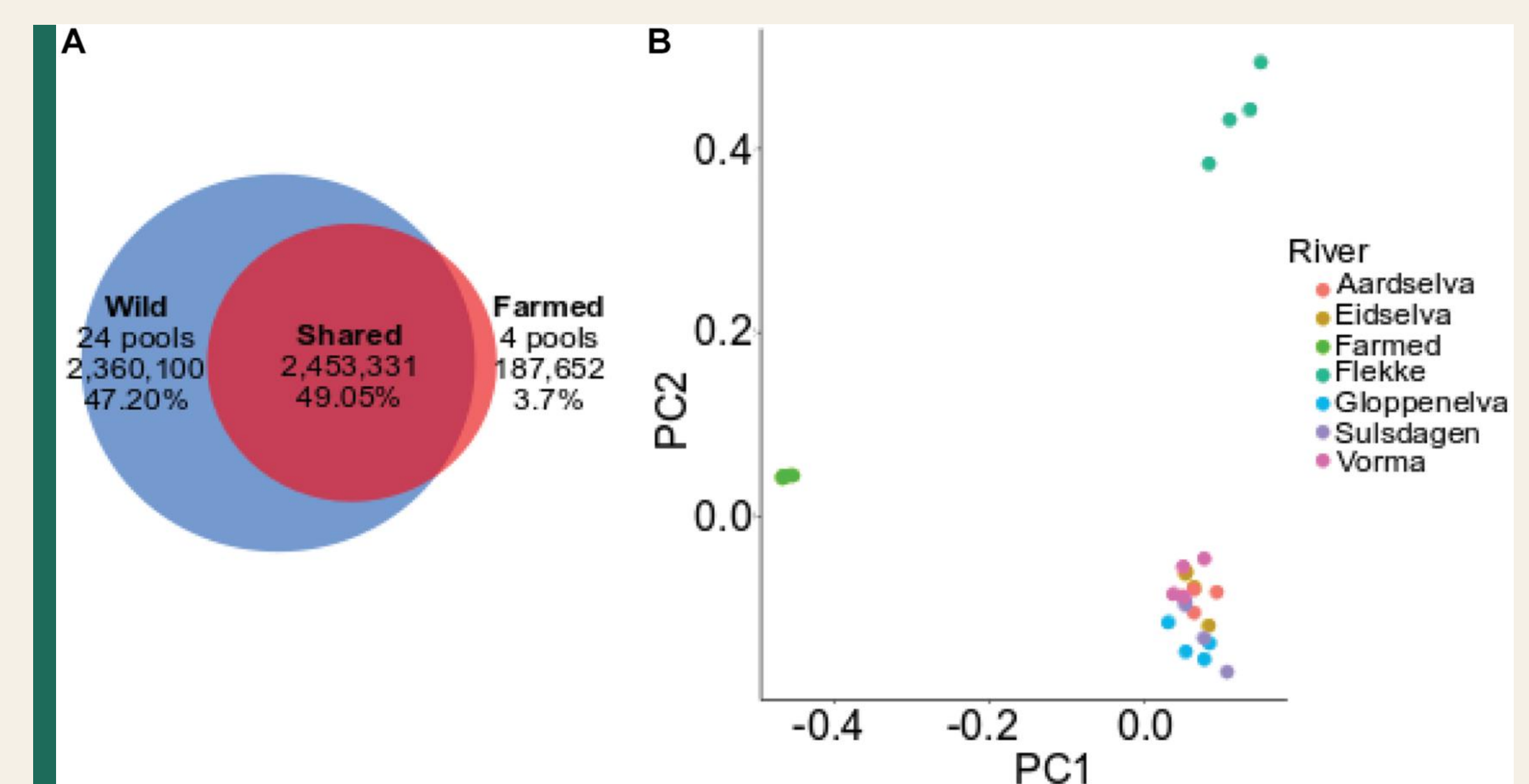


Figure 2. Distribution of pooled heterozygosity (ZHp). (a) Number and proportion of SNP shared between farmed and wild European *Atlantic salmon*. (b) Genomic relationship assessed using PCA of 28 pooled samples; PC1 explains 1.59% and PC2 1.13% of total variability. Extracted from Naval-Sanchez *et al.* (2020).

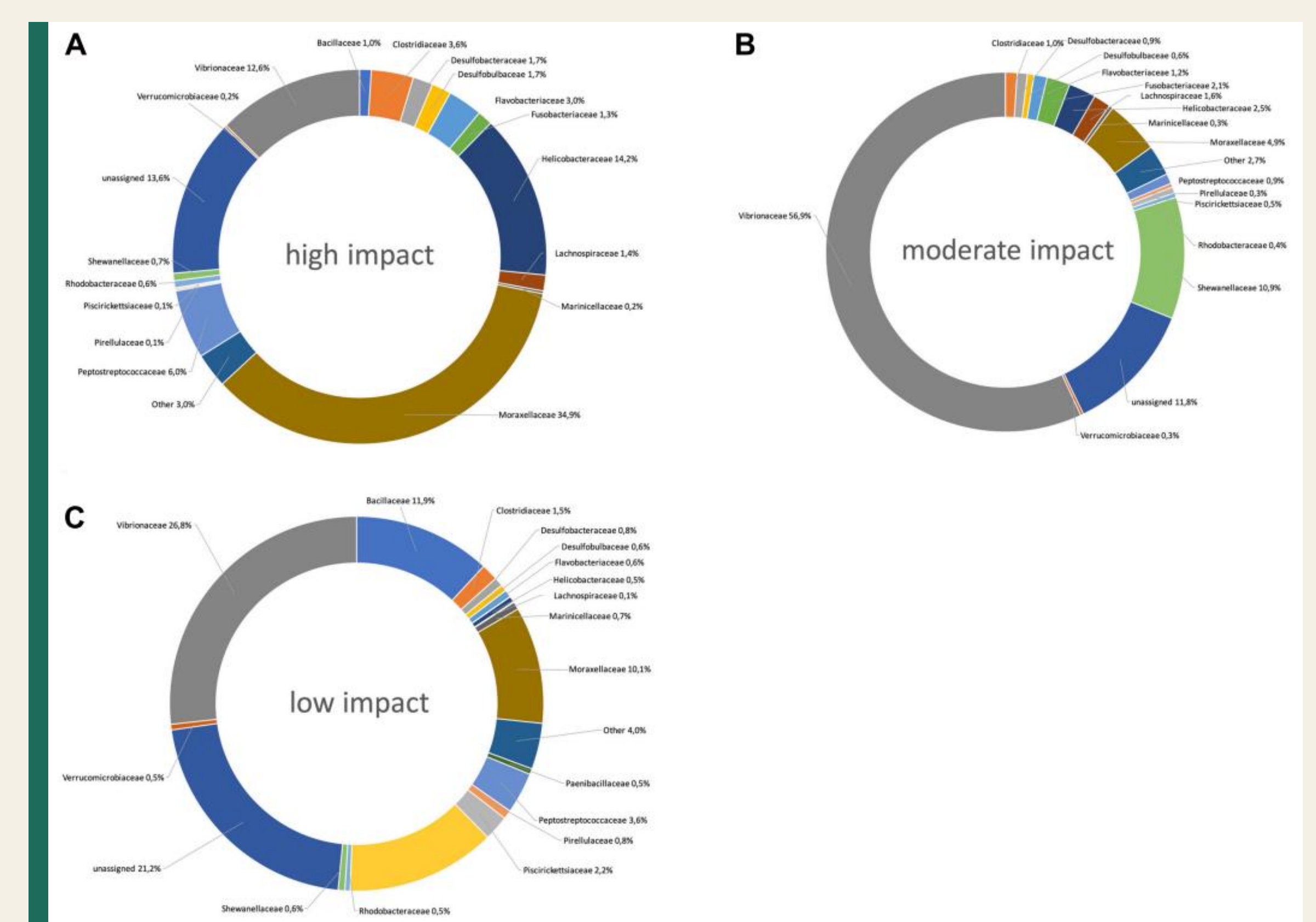


Figure 3. Pie charts showing the taxonomic distribution (relative abundances) of bacterial taxon ranks at high (A), moderate (B), and low (C) aquaculture-related impact sites. Extracted from Frühe *et al.* (2021).

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